CO1 Lab programs:-

(1) **Aim:-** To display a simple line graph using matplotlib.

Pseudocode:-

```
Begin
Import matplotlib
Set x = [1,2,3,4,5]
Set y = [2,4,6,8,10]
Plot x against y as line graph
Display the graph
END
```

Outcome:- A line graph displaying the given x and y values is generated.

(2) **Aim:-** To display the installed version of Scikit-learn.

Pseudocode:-

```
Begin
Import Scikit-learn
Get the installed version
Display the version
End
```

Outcome:- The installed version of Scikit-learn is displayed.

(3) **Aim:-** Create a Numpy Array of the first 20 Natural numbers.

Pseudocode:-

```
Begin
Import the Numpy library
Define the starting number as 1
Define the ending number as 20
Generate natural numbers from 1 to 20
```

Store the numbers in a numpy array
Display the NumPy array
End.

Outcome :- A Numpy array containing natural numbers from 1 to 20 is displayed.

(4) Aim :- To Create a Pandas Dataframe with student names and Grades

Pseudocode :-
Begin
Import the Pandas library
Create a list of student names
Assign names such as Arun etc
Create a list of corresponding Grades.
Assign grades such as A, B and A
Combine the student names and grades
Create a Pandas Dataframe using data
Display the dataframe
End,

Outcome :- A Dataframe containing student names and their corresponding grades is displayed.

(5) Aim: To write a python program to verify the installed versions of commonly used AI Libraries.

Pseudocode:-

```
Begin
    Import package version checking module
    Set libraries = [numpy, Pandas, matplotlib, scikit learn]

    for each library in libraries
        If the library is installed
        Then Get and display its version
        Else
        Display "Not Installed"
        End If
    End for
End
```

Outcome:- The installed versions of the AI libraries are displayed, and missing libraries are identified.

(6) Aim:- To create tensors using PyTorch and Peoform basic mathematical operations.

Pseudocode:-

```
Begin
Import Pytorch
Create tensor A = [2,4,6]
Create tensor B = [1,2,3]
Add tensor A and tensor B
element - wise
Display both results
End.
```

Outcome:- The addition result [3,6,9] and multiplication result [2,8,18] are displayed.

(7) Aim:- To Perform Sentiment analysis using a pre-trained transformer model using Hugging face.

Pseudocode:- Begin
Import the tokenizer, model and Pytorch
Load the pre-trained sentiment analysis model
Input a sentence
Tokenize the sentence
Pass the tokens to the model
Predict the sentiment
Display the sentiment
End.

Outcome:- The Sentiment of the input sentence is classified as positive or negative.

(8) Aim:- To Tokenize a Sentence using the pre-trained Bert Tokenizer.

Pseudo code:- Begin
Import the BERT tokenizer
Load the pre-trained BERT Tokenizer.
Set a sentence as input
Convert the Sentence into tokens.
Display the tokens.
End

Outcome:- The input sentence is divided into tokens, which are displayed.

(9) Aim: Generate text using a pre-trained GPT-2 model.

Pseudocode:- Begin

Import the text-generation pipeline

Load the pre-trained GPT 2 model

Input a starting sentence

Generate additional text using the model

Display the generated text

End.

Outcome: New text is generated based on the given input sentence.

(10) Aim:- To perform sentiment analysis using the hugging face pipeline() API.

Pseudocode: Begin

Import the pipeline function

Create a sentiment analysis pipeline

Input a sentence

Analyse the sentence using the pipeline

Get the sentiment label and confidence sco

End.

Outcome: The sentiment label and its confidence score are displayed.

(11)

Aim:- To load the BERT tokenizer and convert a sentence into tokens.

Pseudocode:-

Begin
Import the Autotokenizer
Load the pre-trained BERT tokenizer
Input a sentence.
Convert the Sentence into tokens
Display the tokens
End

Outcome:- The Sentence is converted into tokens using the BERT tokenizer.

(12) Aim:- To generate Contextual embeddings for a sentence using a pre-trained BERT model.

Pseudocode:-

Begin
Import the BERT tokenizer and model
Load the pre-trained BERT tokenizer and model
Input a sentence
Tokinize the sentence
Pass the tokens through the BERT model
Extract the Contextual embeddings
Display the Embeddings Shape
End.

Outcome:- Contextual embeddings are generated, and their dimensions are displayed.

(13) Aim: To generate text from a given input sentence using GPT-2.

Pseudocode

Begin
Import the text-generation pipeline
Load the pre-trained GPT-2 model
Input a starting sentence
Generate additional text
Display the complete generated text
End.

Outcome: The GPT-2 model generates and displays text based on the given input sentence

(14) Aim: To answer a question based on a given context using a pre-defined transformer model

Pseudocode:

Begin
Import the question-answering pipeline
Load a pre-trained question-answering model
Input a context and a question
Analyse the context and question
Extract the answer from context.
Display the answer
End

Outcome: An answer to the given question is extracted from the provided context and displayed.